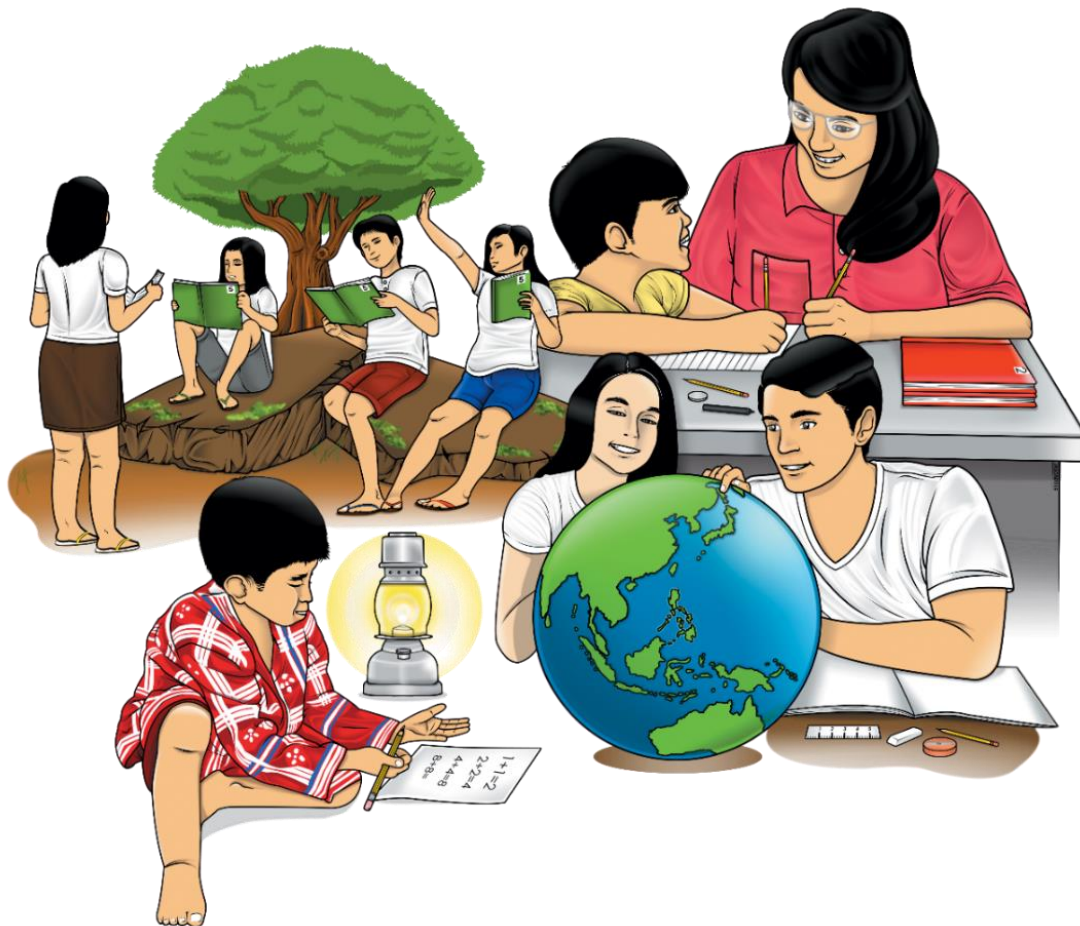


Cellular Respiration and Fermentation

Learner's Module in General Biology 1

Quarter 2. Module 8



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Department of Education. Cordillera Administrative Region

NAME: _____ GRADE AND SECTION: _____
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PREFACE

This module is a project of DepEd Schools Division of Baguio City through the Curriculum Implementation Division (CID) which is in response to the implementation of the K to 12 Curriculum.

This Learning Material is a property of the Department of Education, Schools Division of Baguio City. It aims to improve students' performance specifically Biology.

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Learning Competency	Describe the role of oxygen in respiration and describe pathways of electron flow in the absence of oxygen. Explain the advantages and disadvantages of fermentation and aerobic respiration.

ACKNOWLEDGEMENT

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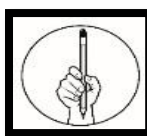


WHAT I NEED TO KNOW

This learning material is intended to provide you with meaningful experiences for independent learning at your own pace and time. You will be supported and guided on the contents of the learning material while you actively participate as a learner.

Learning Objectives:

1. Describe the role of oxygen in respiration and describe pathways of electron flow in the absence of oxygen.
2. Explain the advantages and disadvantages of fermentation and aerobic respiration.



WHAT I KNOW

INSTRUCTION. The following questions will assess your prior knowledge on the lessons that we will take. Choose the letter of the correct answer and write your answers on a separate sheet of paper.

1. Which of the following equations accurately represents cellular respiration?
 - a. $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + \text{energy}$
 - b. $6CO_2 + 6H_2O \longrightarrow C_6H_{12}O_6 + 6O_2 + \text{energy}$
 - c. $6CO_2 + 6O_2 + \text{energy} \longrightarrow C_6H_{12}O_6 + 6H_2O$
 - d. $C_6H_{12}O_6 + 6H_2O + \text{energy} \longrightarrow 6CO_2 + 6O_2$
2. ATP is an important biological molecule that transports chemical energy within cells and powers many biochemical reactions. Which of the following correctly describes the structure of ATP?
 - a. ATP contains a 5-carbon sugar, three phosphate groups, and an adenine base.
 - b. ATP contains a 6-carbon sugar, two phosphate group, and an adenine base.
 - c. ATP contains a 5-carbon sugar, two phosphate groups, and an adenine base.
 - d. ATP contains a 6-carbon sugar, three phosphate groups and an adenine base.
3. Which of the following statements is TRUE about glycolysis?
 - a. It can occur without the presence of oxygen.
 - b. It takes place in the mitochondria of a cell.
 - c. It is the final step in aerobic respiration.
 - d. It produces a net of 1 ATP, 1 NADH, and 2 pyruvate.
4. Energy is stored long-term in the bonds of ____ and is used short-term to perform work from an ____ molecule.
 - a. ATP: glucose
 - b. an anabolic molecule: catabolic molecule
 - c. glucose: ATP
 - d. a catabolic molecule: anabolic molecule

5. What type of organisms live in the absence of oxygen?
 - a. aerobes
 - b. facultative anaerobes
 - c. anaerobes
 - d. obligate anaerobes
6. These organisms live with the presence of oxygen but still does not use it?
 - a. aerobes
 - b. facultative anaerobes
 - c. anaerobes
 - d. obligate anaerobes
7. The glucose that enters the glycolysis pathway is split into two molecules of _____.
 - a. ATP
 - b. NADH
 - c. phosphate
 - d. pyruvate
8. What would happen to the flow of electrons if oxygen were not present?
 - a. Electrons continue to be transported.
 - b. The ETC can run by itself without the presence of oxygen.
 - c. The ETC cannot run because there is no oxygen to act as the final acceptor.
 - d. The presence of oxygen is not necessary as electrons are transported.
9. What do the electrons added to NAD^+ do?
 - a. They become part of a fermentation pathway.
 - b. They go to another pathway for ATP production.
 - c. They energize the entry of the acetyl group into the citric acid cycle.
 - d. They are converted into NADP.
10. Which of the following occur during the Krebs cycle?
 - a. Acetyl-CoA combines with oxaloacetate to form citrate.
 - b. FADH_2 accepts two electrons in order to form FAD.
 - c. NAD^+ is reduced to form NADH
 - d. Pyruvate is converted into acetyl-CoA.
11. The electron transport chain (ETC) is one part in oxidative phosphorylation ,which is the last step in cellular respiration. Which of the following is TRUE about the electron transport chain?
 - a. The movement of electrons down the ETC drives the formation of a proton gradient.
 - b. The movement of H^+ ions across the inner membrane of the mitochondria drives the transfer of electrons down the ETC.
 - c. The movement of H^+ ions through ATP synthase drives the transfer of electrons down the ETC.
 - d. The movement of electrons down the ETC drives the movement of ADP and phosphate through ATP synthase.
12. Which of the following fermentation methods can occur in animal skeletal muscles?
 - a. alcohol fermentation
 - b. lactic acid fermentation
 - c. mixed acid fermentation
 - d. propionic fermentation

13. How do lactic acid and alcohol fermentation differ?
- Carbon dioxide is released only during alcohol fermentation.
 - Lactic acid fermentation produces lactic acid as a byproduct and alcohol fermentation produces ethanol.
 - Lactic acid fermentation can occur in large muscles in the human body, while alcohol fermentation occurs in yeast and some bacteria.
 - Lactic acid fermentation produces a 2-carbon compound, while alcohol fermentation produces a 3-carbon compound
14. Frank is investigating how exercise affects the rate at which he can squeeze a ball. After several minutes, he begins to experience muscle soreness and fatigue. He knows that this means his muscles are beginning to run out of oxygen. Which process is likely causing the soreness in her muscle cells?
- alcohol fermentation
 - Krebs cycle
 - Lactic Acid Fermentation
 - photosynthesis
15. In the absence of oxygen, some organisms produce ethanol (alcohol) and ATP from pyruvate and NADH. What anaerobic pathway is involved?
- Alcohol fermentation
 - Glycolysis
 - Krebs cycle
 - Lactic acid fermentation

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WHAT'S IN

Let's have a review on what you have learned from the previous lesson.

Name: _____
Strand and Grade Level: _____

Date: _____
Score: _____/5

In the previous module, you have learned the features of glycolysis, Krebs Cycle, electron transport system and chemiosmosis. Further, you have described the reaction that produce and consume Adenosine Triphosphate (ATP). Those lessons will help you describe the role of oxygen in respiration, the pathways of electron in the absence of oxygen, and the advantages and disadvantages of fermentation and aerobic respiration.



WHAT'S NEW?

ACTIVITY 1. FACT OR BLUFF

Write FACT if the statement correctly describes the features of the three stages of cellular respiration and BLUFF if otherwise. Write your answers on a separate sheet of paper.

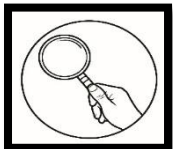
_____ 1. Glycolysis is the process that breaks down one molecule of glucose into two 3-carbon molecules of pyruvate or pyruvic acid.

_____ 2. The Krebs cycle is a cyclical series of enzyme-controlled reactions that occurs in the matrix of the mitochondria of a cell.

_____ 3. Krebs cycle is sometimes called citric acid cycle (CAC)

_____ 4. The electron transport chain (ETC) is a series of proton pumps on the outer membrane of the mitochondrion.

_____ 5. Electron transport is the last stage of cellular respiration.



WHAT IS IT?

How Is Oxygen Important to the Release of Energy in Cellular Respiration?

Cellular respiration is the process by which cells release energy from glucose and change it into a usable form called ATP (adenosine triphosphate). ATP is a molecule that provides a small amount of energy to the cell, which provides it fuel to do specific tasks.

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There are two types of respiration: **anaerobic and aerobic**. In most cells, when oxygen is present, they carry out aerobic respiration. **Aerobic cellular respiration** is the process by which cells use oxygen to help them convert glucose into energy.

Aerobic respiration occurs in three stages where a glucose molecule is the source of energy. The first stage is called **glycolysis** and does not require oxygen. In this stage, ATP molecules are used to help break down glucose into a substance called pyruvate, a molecule that transports electrons called NADH, two more ATP molecules, and carbon dioxide. Carbon dioxide is a waste product and is removed from the body. The second stage is called the **Krebs cycle**. This cycle consists of a series of complex chemical reactions that generate additional NADH. The final stage is called **electron transport phosphorylation**. During this stage, NADH and another transporter molecule called FADH₂ carry electrons to the cells. Energy from the electrons is converted to ATP. Once the electrons have been used, they are donated to atoms of hydrogen and oxygen to make water.

Aerobic respiration is more efficient and it yields a lot of ATP. However, there are many organisms in this world that thrive in mud, marshes, animal gut, canned goods, sewage treatment pond, and deep oceans where oxygen is scarce. Organisms that live in the absence of oxygen are called **anaerobes**. Some actually die if exposed to oxygen. They are called **obligate anaerobes**. On the other hand, some can live with the presence of oxygen but still does not use it. They are called **facultative anaerobes**.

Cellular respiration that proceeds without the presence of oxygen is called **anaerobic respiration**. Anaerobic respiration produces yeast or lactate. When exercising, the body uses oxygen more quickly than it is taken in; anaerobic respiration provides lactate to keep the muscles moving. Lactate build up and lack of oxygen are the reasons for muscle fatigue and labored breathing during hard exercise.

During **anaerobic respiration**, only glycolysis occurs. The 2 molecules of NADH that are generated during glycolysis are then converted back into NAD⁺ during anaerobic respiration so that glycolysis can continue. Since glycolysis only produces 2 ATP, anaerobic respiration is much less efficient than aerobic respiration (2 ATP molecules compared to 36 ATP molecules). However, 2 ATP molecules is much better for a cell than ATP molecules. In anaerobic situations, the cell needs to continue performing glycolysis to generate 2 ATP per glucose because if a cell is not generating any ATP, it will die.

Note that the only part of aerobic respiration that physically uses oxygen is the electron transport chain. However, the citric acid cycle cannot occur in the absence of oxygen because there is no way to regenerate the NAD⁺ used during this process.

In the event that the oxygen supply in aerobic cells becomes low, they also do anaerobic pathways. There are two common anaerobic pathways happen in these cells: **alcoholic fermentation and lactate fermentation**.

Alcohol Fermentation

Another familiar fermentation process is alcohol fermentation (**Figure 2**), which produces ethanol, an alcohol. The alcohol fermentation reaction is the following:

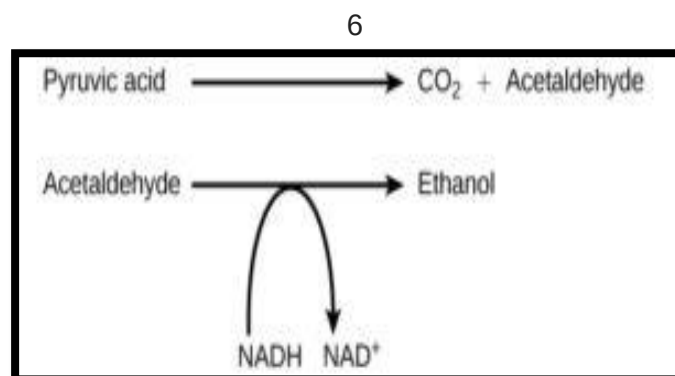


Figure 2. The reaction resulting in alcohol fermentation is shown.
Image taken from www.rses.lumenlearning.com

The fermentation of pyruvic acid by yeast produces the ethanol found in alcoholic beverages. If the carbon dioxide produced by the reaction is not vented from the fermentation chamber, for example in beer and sparkling wines, it remains dissolved in the medium until the pressure is released. Ethanol above 12 percent is toxic to yeast, so natural levels of alcohol in wine occur at a maximum of 12 percent. Again, the purpose of this process is not to produce ethanol, but rather to convert NADH back into NAD^+ so that glycolysis can continue.

In **alcoholic fermentation**, ethyl alcohol and carbon dioxide are produced by some cells using the pyruvate from glycolysis. Each pyruvate molecule is rearranged into acetaldehyde and carbon dioxide is released. NADH gives up electrons to acetaldehyde and form the end product ethanol. Fermentation is widely used in the industry.

Yeast is a fungi used in making some bread and can undergo anaerobic respiration. Bakers mix the sugar, flour, water, and yeast to form the bread dough. The dough rises due to the carbon dioxide released by the yeast cells. Beer and wine manufacturers use yeast to ferment the sugars in wheat and grape juice forming alcoholic beverages such as beer, wine, etc.

In some cells, glycolysis produces two pyruvate and two NADH molecules and two ATP. **Pyruvate** itself becomes the final acceptor of the electrons from the NADH that produces the final product, **lactate**. Often times, this product is called lactic acid. Human skeletal muscles can carry out fermentation when the blood cannot supply the cells with adequate oxygen during some strenuous activities. Lactic acid as product of fermentation will build up in the muscle that causes fatigue, burning sensation, and cramps. Lactic acid will continue to build up until there is adequate supply of oxygen. Lactic acid is then converted back into pyruvate in the liver. Normal functioning of the muscles will also happen.

It is important to note that fermentation pathways do not breakdown and utilize the glucose completely. ATP is no longer produced beyond the process of glycolysis. Thus, energy produced is just enough for some single-celled organisms or can only sustain energy for multicellular organisms for a short period of time.

Glycolysis and the citric acid cycle cannot occur if there is no NAD^+ present to pick up electrons as the reactions proceed. When oxygen is present, this isn't a problem – all of the NADH and FADH_2 that were produced during glycolysis and the citric acid cycle are converted back into NAD^+ and FAD after the electron transport chain.

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When no oxygen is present, the electron transport chain can't run because there is no oxygen to act as the final electron acceptor. This means that the ETC will not be accepting electrons from NADH as its source of power, so NAD^+ will not be regenerated. Both glycolysis and the citric acid cycle require NAD^+ to accept electrons during their chemical reactions. In order for the cell to continue to generate any ATP, NADH must be converted back to NAD^+ for use as an electron carrier. Anaerobic processes use different mechanisms, but all function to convert NAD^+ back into NADH.

How is this done?

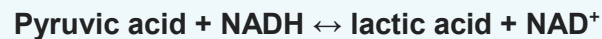
- Processes that use an organic molecule to regenerate NAD^+ from NADH are collectively referred to as **fermentation**.

- In contrast, some living systems use an inorganic molecule (such as nitrate or sulfur) to regenerate NAD^+ .

Both of these methods are called **anaerobic cellular respiration**. They do not require oxygen to achieve NAD^+ regeneration and enable organisms to convert energy for their use in the absence of oxygen.

Lactic Acid Fermentation

The fermentation method used by animals and some bacteria like those in yogurt is lactic acid fermentation (**Figure 1**). This occurs routinely in mammalian red blood cells and in skeletal muscle that does not have enough oxygen to allow aerobic respiration to continue (such as in muscles after hard exercise). The chemical reaction of lactic acid fermentation is the following:



The build-up of lactic acid causes muscle stiffness and fatigue. In muscles, lactic acid produced by fermentation must be removed by the blood circulation and brought to the liver for further metabolism. Once the lactic acid has been removed from the muscle and is circulated to the liver, it can be converted back to pyruvic acid and further catabolized (broken down) for energy.

Note that the purpose of this process is not to produce lactic acid (which is a waste product and is excreted from the body). The purpose is to convert NADH back into NAD^+ so that glycolysis can continue so that the cell can produce 2 ATP per glucose.

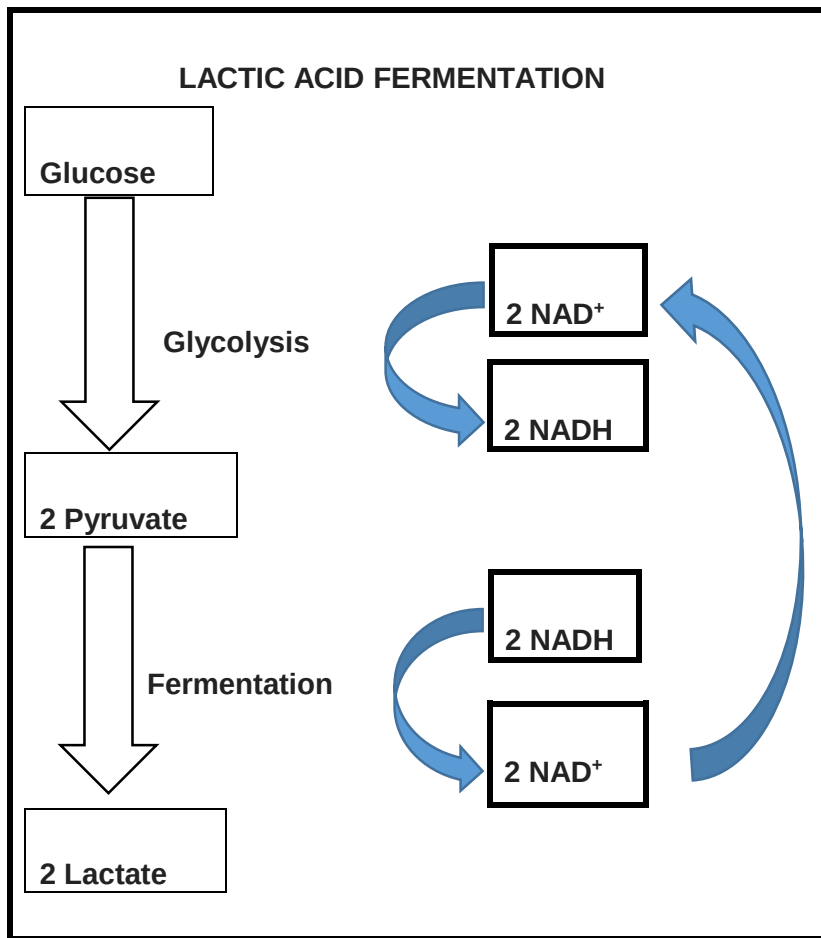


Figure 1. Lactic acid fermentation is common in muscles that have become exhausted by use.
Image taken from www.rses.lumenlearning.com

To sum up, if NADH cannot be metabolized through aerobic respiration, another electron acceptor is used.

Most organisms will use some form of fermentation to accomplish the regeneration of NAD⁺, ensuring the continuation of glycolysis. The regeneration of NAD⁺ in fermentation is not accompanied by ATP production; therefore, the potential for NADH to produce ATP using an electron transport chain is not utilized.

How does the pathway of electron and energy production change in the absence of oxygen?

Aerobic respiration requires oxygen because oxygen is the final electron acceptor of the electron transport chain in the mitochondria. If there is no oxygen to accept electrons, then the electron transport chain stops working and the high energy molecules NADH and FADH₂ cannot be converted back into NAD and FAD.

When oxygen is not present what pathway occurs?

Glycolysis, which is the first step in all types of cellular respiration is anaerobic and does not require oxygen. If oxygen is present, the pathway will continue on to the Krebs cycle and oxidative phosphorylation. However, if oxygen is not present, some organisms can undergo fermentation to continually produce ATP.



WHAT'S MORE?

ACTIVITY 1

INSTRUCTION: Rearrange the letters on the LEFT COLUMN to reveal the words described on the RIGHT COLUMN. Write your answers on a separate sheet of paper.

1. bicroanae	cellular respiration that proceeds without the presence of oxygen
2. boricae	cells use oxygen to help them convert glucose into energy
3. colyglysis	first stage of aerobic respiration
4. phoryphostionla	final stage of aerobic respiration
5. libotega	they die if exposed to oxygen
6. taculfative	they can live with the presence of oxygen but still does not use it
7. reftamention	uses an organic molecule to regenerate NAD ⁺ from NADH
8. vapyteru	final acceptor of the electrons from the NADH
9. tacalte	final product
10. berks cleyc	second stage of aerobic respiration

ASSESSMENT 1. TRUE OR FALSE

INSTRUCTION. Based on the above activity, answer the following statements on anaerobic and aerobic respiration. Write **TRUE** if the statement is correct; **FALSE** if otherwise. Write your answers on a separate sheet of paper. (10 points)

- _____ 1. Aerobic respiration requires oxygen because oxygen is the final electron acceptor of the electron transport chain in the mitochondria. T
- _____ 2. Glycolysis, which is the first step in all types of cellular respiration is anaerobic and does not require oxygen. T
- _____ 3. Most organisms will use some form of fermentation to accomplish the regeneration of NAD⁺, ensuring the continuation of glycolysis. T
- _____ 4. The build-up of lactic acid causes muscle stiffness and fatigue. T
- _____ 5. The fermentation method used by animals and some bacteria like those in yogurt is alcohol fermentation. F
- _____ 6. Processes that use an organic molecule to regenerate NAD⁺ from NADH are collectively referred to as fermentation. T
- _____ 7. In some cells, glycolysis produces two pyruvate and two NADH molecules and three ATP. F
- _____ 8. In lactic acid fermentation, ethyl alcohol and carbon dioxide are produced by some cells using the pyruvate from glycolysis. F
- _____ 9. Cellular respiration that proceeds without the presence of oxygen is called anaerobic respiration. T
- _____ 10. Cellular respiration is the process by which cells release energy from glucose and change it into a usable form called ATP. T

ACTIVITY 2. FIND ME

INSTRUCTION. Find the 10 words below that are related to cellular respiration. The words may appear straight across, backward straight across, up and down. Write your answers on a separate sheet of paper.

A	B	C	D	N	A	D	E	N	O	S	I	N
G	F	E	E	I	W	R	D	Y	T	U	I	K
H	I	N	S	C	W	F	E	G	T	Y	U	I
E	D	I	X	O	I	D	N	O	B	R	A	C
D	D	S	P	T	D	D	E	E	F	R	D	D
I	A	O	Y	I	I	P	H	Y	H	G	K	I
T	S	N	R	N	C	E	T	H	A	N	O	L
O	D	E	U	A	A	A	D	E	R	T	P	H
E	U	D	V	M	C	P	Y	R	U	V	A	T
L	N	A	A	I	I	E	N	I	N	E	D	A
C	M	D	T	D	T	F	A	R	S	T	U	B
U	V	A	E	E	C	A	S	D	E	T	R	G
N	A	C	E	T	A	L	D	E	H	Y	D	E
I	S	D	A	S	L	E	R	T	Y	U	N	D
D	E	T	A	H	P	S	O	H	P	I	R	T

ASSESSMENT 2

INSTRUCTION: Answer the questions concisely. Use a separate sheet of paper to write your answer. (15 points)

1. Explain the role of oxygen in respiration.

2. Describe the two anaerobic pathways.

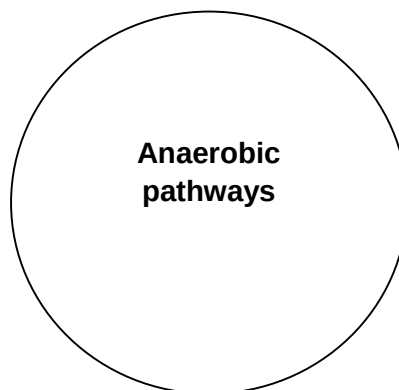
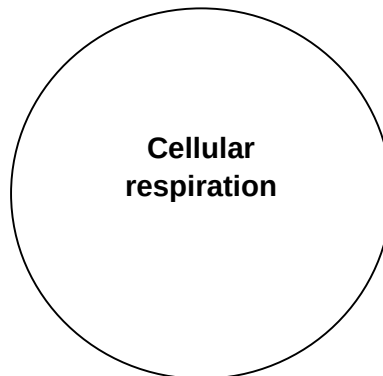
3. Fermentation is widely used in the industry. Enumerate at least three advantages and disadvantages.



WHAT I HAVE LEARNED

Activity: Summary

Using the concepts learned, make a **MIND MAP** starting from the word/phrase inside the circles. Use a separate sheet of paper to write your answers. (10 points)





WHAT I CAN DO

APPLICATION:

Having learned the lesson on cellular respiration and fermentation, you are now ready to explore and apply in real-life situations.

INSTRUCTION. Do the activity below by following the step-by-step procedures.

Title of the Activity: Ferment Your Own Food

1. Prepare the following equipment/materials:
 - air-tight lid jars or bottle container
 - 33 grams of salt or one tablespoon of salt
 - 4 cups of clean water
 - fresh cucumbers
2. Wash properly the jar or bottle container to be used.
3. Wash the fresh cucumbers.
4. Soak cucumbers in an ice water bath for a couple of hours. Submerging cucumbers in an icy bowl of water in the fridge will help them firm up/stay firm.
5. Cut off the blossom end of cucumber- where it was attached to the vine.
6. Put the 4 cups of water inside the jar or bottle container. Put also the tablespoon of salt.
7. Carefully slice the cucumbers in small cut. Put inside the jar or bottle container.
8. Close the lid tightly and put in a clean area for 2 and 1/2 days to 4 days to be perfect.
9. Your fermented food is now ready to be eaten.
10. Take a photo of what you have prepared and send to your teacher for recording purposes.



ASSESSMENT

INSTRUCTION. The following questions will assess how well you were able to grasp and understand concepts on cellular respiration and fermentation. Choose the correct answer from the choices given and write on a separate sheet of paper.

1. Which of the following equations accurately represents cellular respiration?
 - a. $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \longrightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy}$
 - b. $6\text{CO}_2 + 6\text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + \text{energy}$
 - c. $6\text{CO}_2 + 6\text{O}_2 + \text{energy} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O}$
 - d. $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + \text{energy} \longrightarrow 6\text{CO}_2 + 6\text{O}_2$
2. Energy is stored long-term in the bonds of ____ and is used short-term to perform work from an ____ molecule.
 - a. ATP: glucose
 - b. an anabolic molecule: catabolic molecule
 - c. glucose: ATP
 - d. a catabolic molecule: anabolic molecule
3. Which of the following statements is TRUE about glycolysis?
 - a. It can occur without the presence of oxygen.
 - b. It takes place in the mitochondria of a cell.
 - c. It is the final step in aerobic respiration.
 - d. It produces a net of 1 ATP, 1 NADH, and 2 pyruvate.
4. These organisms live with the presence of oxygen but still does not use it?
 - a. aerobes
 - b. facultative anaerobes
 - c. anaerobes
 - d. obligate anaerobes
5. The glucose that enters the glycolysis pathway is split into two molecules of _____.
 - a. ATP
 - b. NADH
 - c. phosphate
 - d. pyruvate
6. ATP is an important biological molecule that transports chemical energy within cells and powers many biochemical reactions. Which of the following correctly describes the structure of ATP?
 - a. ATP contains a 5-carbon sugar, three phosphate groups, and an adenine base.
 - b. ATP contains a 6-carbon sugar, two phosphate group, and an adenine base.
 - c. ATP contains a 5-carbon sugar, two phosphate groups, and an adenine base.
 - d. ATP contains a 6-carbon sugar, three phosphate groups and an adenine base.
7. What type of organisms live in the absence of oxygen?
 - a. aerobes
 - b. facultative anaerobes
 - c. anaerobes
 - d. obligate anaerobes
8. What would happen to the flow of electrons if oxygen were not present?
 - a. Electrons continue to be transported.
 - b. The ETC can run by itself without the presence of oxygen.
 - c. The ETC cannot run because there is no oxygen to act as the final acceptor.
 - d. The presence of oxygen is not necessary as electrons are transported.

9. Which of the following occur during the Krebs cycle?
- Acetyl-CoA combines with oxaloacetate to form citrate.
 - FADH₂ accepts two electrons in order to form FAD.
 - NAD⁺ is reduced to form NADH
 - Pyruvate is converted into acetyl-CoA.
10. The electron transport chain (ETC) is one part in oxidative phosphorylation ,which is the last step in cellular respiration. Which of the following is TRUE about the electron transport chain?
- The movement of electrons down the ETC drives the formation of a proton gradient.
 - The movement of H⁺ ions across the inner membrane of the mitochondria drives the transfer of electrons down the ETC.
 - The movement of H⁺ ions through ATP synthase drives the transfer of electrons down the ETC.
 - The movement of electrons down the ETC drives the movement of ADP and phosphate through ATP synthase.
11. What do the electrons added to NAD⁺ do?
- They become part of a fermentation pathway.
 - They go to another pathway for ATP production.
 - They energize the entry of the acetyl group into the citric acid cycle.
 - They are converted into NADP.
12. Which of the following fermentation methods can occur in animal skeletal muscles?
- alcohol fermentation
 - lactic acid fermentation
 - mixed acid fermentation
 - propionic fermentation
13. In the absence of oxygen, some organisms produce ethanol (alcohol) and ATP from pyruvate and NADH. What anaerobic pathway is involved?
- Alcohol fermentation
 - Glycolysis
 - Krebs cycle
 - Lactic acid fermentation
14. How do lactic acid and alcohol fermentation differ?
- Carbon dioxide is released only during alcohol fermentation.
 - Lactic acid fermentation produces lactic acid as a byproduct and alcohol fermentation produces ethanol.
 - Lactic acid fermentation can occur in large muscles in the human body, while alcohol fermentation occurs in yeast and some bacteria.
 - Lactic acid fermentation produces a 2-carbon compound, while alcohol fermentation produces a 3-carbon compound
15. Frank is investigating how exercise affects the rate at which he can squeeze a ball. After several minutes, he begins to experience muscle soreness and fatigue. He knows that this means his muscles are beginning to run out of oxygen. Which process is likely causing the soreness in her muscle cells?
- alcohol fermentation
 - Krebs cycle
 - Lactic Acid Fermentation
 - photosynthesis

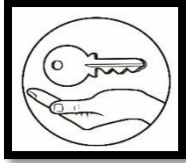


ADDITIONAL ACTIVITY

INSTRUCTION: Answer the question concisely. Use a separate sheet of paper to write your answer.

1. Why milk or creams turn sour after sometime?

Education is a trail to be freed from societal suppression” pbl



ANSWEK KEY

Pre-Assessment	1. A	1. anaerobic	1. A	1. A
	2. A	2. aerobic	2. C	2. C
	3. A	3. glycolysis	3. A	3. A
	4. C	4.	4. B	4. B
	5. C	5. obligate	5. D	5. D
	6. B	6. facultative	6. A	6. A
	7. D	7. fermentation	7. C	7. C
	8. B	8. pyruvate	8. B	8. B
	9. C	9. lactate	9. A/C	9. A/C
	10. A/C	10. glucose	10. A	10. A
	11. A	Assessment 1	11. C	11. C
	12. B	1. T	12. B	12. B
	13. A/B/C	2. T	13. A	13. A
	14. C	3. T	14. A/B/C	14. A/B/C
	15. A	4. T	15. C	15. C
		5. F		
		6. T		
		7. F		
		8. F		
		9. T		
		10. T		

Activity 1	1. acetalddehyde	Assessment 2	Answers may vary according to the student's perception
	2. ethanol		
	3. carbon dioxide		
	4. dinucleotide		
	5. adenine		
	6. adenosine		
	7. pyruvate		
	8. lactic acid		
	9. nicotinamide		
	10. triphosphate		

Activity 2	1. A	Assessment	
	2. C		
	3. A		
	4. B		
	5. D		
	6. A		
	7. C		
	8. B		
	9. A/C		
	10. A		
	11. C		
	12. B		
	13. A		
	14. A/B/C		
	15. C		

REFERENCES

- Bear, R., et. al. 2016. "Overview of Cellular Respiration." Principle of Biology. OpenStax CNX. http://cnx.org/contents/24nl-KJ8@24.18:R_v3DfP5@9/Overview-of-cellular-Respirati.
- Rave, P.H.,et. al. 2014. "How Cells Harvest Energy." In Biology, 122-46. 10th ed. AP Edition. New York, NY: McGraw Hill.
- Reece, J.B. et. al. 2011. "Cellular Respiration and Fermentation." In Campbell Biology, 162-84. 10th ed. San Francisco, CA: Pearson.
- Ori, J., 2018. Steps of Cellular Respiration. Khan Academy. Retrieved from: www.sciencing.com/oxygen-respiration
- Sia, R. D. and Cortez, L. A. S., 2016. Science in Today's World for Senior High School. Earth and Life Science. Sibs Publishing House, Inc.
- Retrieved from:http://commons.wikimedia.org/wiki/File:Blausen_0770_Respiratory-System_02.png;
- Retrieved from:<https://www.khanacademy.org/science/biology/cellular-molecular-biology/cellular-respiration/v/oxidation-and-reduction-from-biological-view? v=orl2m6larJg>.
- Retrieved from: OpenStax. <http://cnx.org/contents/b3c1d2-839c-42b0-a314-e119a8aafbdd@8.10:1/Concepts of Biology>
- Retrieved from: <http://www.savorylotus.com/easy-fermented-cucumbers-pickles/>"respond"

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